

PERSEIDS captures audio into up to five Trails. From those it builds a stereo sound cloud: Spectra (stylized additive) blended with Swarm (grains), plus Resonator, Filter, Reverb and spatial motion. Multi Dry/Wet balances clean input against the finished cloud. Full wet = cloud only.



Panel layout — labels in the photo, block and pot number in the legend.

WHAT SITS WHERE

- **Trails** — Block 1 · pot 1
- **Time** — Block 2 · pot 2
- **Engines** — Block 3 · pot 3
- **Swarm** — Block 5 · pot 4 — yellow cap
- **Spectra** — Block 4 · pot 5 — green cap
- **PanDrift** — Block 8 · pot 6
- **Reson.** — Block 7 · pot 7
- **Reverb** — Block 6 · pot 8
- **x-Fade** — Block 9 · pot 9
- **Filter** — Block 10 · pot 10
- **T1 - T5** — Trail encoders with a push. Turn = level, short = Lock or back to Home, long = Solo.
- **Multi** — Block 11 · big encoder: D/W · M1 · M2 · TU · SET.
- **Rec** — Manual capture — the same as a trigger on the Trig jack.
- **Play/Pause** — Imprint button: short = Play/Pause, long = lock all active Trails.
- **Cycle** — Button next to the display: hold + turn = scroll the list, long = delete all Trails.
- **in L/R · out L/R** — Audio I/O, mono-cable friendly. Sidechain mode (Settings) splits In L (dry, not recorded) from In R (capture only).

OPERATION — GESTURES

- **Turn a block pot** — Opens that block's menu and edits the last-bound parameter. The value never jumps — it is picked up once the pot has caught it. From another open block, a decisive turn on a different pot jumps straight there.
- **Cycle hold + pot / Multi turn** — Scrolls the parameter list without changing values. On release the pot is bound to the last parameter shown.
- **Cycle short** — While Multi/Settings is open: next entry (stand-in for the unreliable Multi push). Otherwise unused on its own — except to confirm a delete.
- **Cycle long** — “Delete all Trails?” — a short press within 3 s confirms and clears every Trail. Timeout or moving a control cancels.
- **Trail encoder short** — On Home: Lock (L). In a block/Multi menu: back to Home without locking — so from a menu it takes two presses to reach Lock.
- **Trail encoder long** — Solo (S), from any screen. Solo overrides the Crossfade wave.
- **Imprint short** — Play/Pause, global across all Trails, using the fade times from block 2.
- **Imprint long (≈800 ms)** — Locks every active Trail at once — freezes the moment without the gap five separate encoder presses would leave. Toggle: a second long press releases only the Imprint locks, manual locks stay.
- **Multi turn** — Opens the Multi menu (Dry/Wet by default) and edits the bound entry. The Multi push is not live on the current prototype — stepping runs via Cycle until it is.
- **Rec** — Manual capture — the same overwrite rules apply as for the Threshold trigger.
- **Idle ≈ 7 s** — With no pot or button activity the display returns to Home. Only the view changes; pot bindings stay as they were.

READING THE OLED

- Header: block name on the left, position n/m on the right, plus the load meter when enabled.
- Value centred above the bars, abbreviations in the bottom segmented row, active cell inverted. Columns always use the width of a 4-slot page — with 2-3 params space is left empty on the right, with more than four the row scrolls.
- Pole hints beside the value: SP/SW (Blend), SA/FO (Waveshape), UM/AU (Umbra/Aurora), BL/RD (Atmosphere), CH/FR (Character). Below 50 % the left pole sits before the number, above it the right pole after, at centre neither. While a hint shows, the % sign is dropped. Pitch and Velocity stay plain ±%.
- **L!** in the header means the module is temporarily running fewer Swarm grains so playback does not break up. The cloud sounds slightly thinner meanwhile; the marker clears itself once there is headroom again.
- R1...R5 / REC1...REC5 in the header centre show which slot is up — idle, or armed and recording.

The Settings block has no pot of its own — it lives inside the Multi encoder under SET.

BLOCK POTS 1-10

1 TrailsPot 1

How many voices live in the cloud, and when recording starts.

- CNT

Count — 1-5 Trails for this session. Boot default from Settings #T (usually 3). Fewer = thinner, more = denser stack. The dashboard shows exactly this many rows.
- THR

Threshold — How loud the input must be for auto-record. Lower = more sensitive. The value is drawn as a marker right on the VU meter.
- CRE

Cont. Rec — OFF = rising edge only (below → above THR). ON = keep re-arming while the signal stays above the threshold.
- OVR

Overwrite — ON (default) = an empty slot first, otherwise the oldest unlocked Trail is replaced even mid-Hold — faded out softly, so it never clicks. OFF = wait for a free slot; running Trails are protected, only INF stays replaceable. Applies to Threshold, Cont. Rec and Rec.
- ON

On/Off — Capture system on or off.

Only one Trail records at a time — a new trigger only takes hold once the running take is finished. Lock beats everything: a locked Trail is neither overwritten nor faded out.

2 TimePot 2

How long each take is, how long it stays, how it breathes.

- BUF

Buffer — Max recording length per Trail, up to 30 s (boot 2 s). Short = tight loops, long = wide phrases. The loop seam is blended automatically, so the wrap stays seamless.
- HLD

Hold — How long a Trail stays. Boot 15 s, far right → INF. The remaining time runs as a countdown inside the Life-Bar.
- FIN

Fade In — Soft onset of every Trail, up to 5 s (boot 3 s) — also the Play fade-in.
- FOUT

Fade Out — Soft tail before Empty, up to 5 s (boot 3 s) — also the Pause fade-out.

Time Unit (seconds ↔ clock) is one shared global setting for BUF and HLD and lives in Multi — still a dummy slot today.

3 EnginesPot 3

Balance between the two engines and their pitches.

- BLD

Blend — 0 % = pure Spectra, 100 % = pure Swarm, both at once in between with no level dip. Header SP/SW; the centre carries only a subtle hint, because it means “equal mix”, not “no effect”.
- PSP

Pitch Spectra — ±1 octave on the additive body. Centre = original.
- PSW

Pitch Swarm — ±1 octave on the grains.

3 · Engines (continued)

- PB

Pitch Both — New: not a pitch of its own, but the span. Expands PSP and PSW from ±1 oct (PB 0) to ±2 oct (PB 100). Pot travel stays the same, the readout scales with it. Pitched far up it gets gritty.

4 SpectraPot 5

Stylized harmonic body — character resynthesis, not a transparent freeze.

- PAR

Partials — 4-32 partials (default 16). Fewer = simpler, more = denser harmonics.
- WSH

Waveshape — Centre = sine, left toward saw (header SA), right toward fold (FO).
- UMB

Umbra / Aurora — Left (UM) cuts fundamentals and brings quiet noise components forward — airy, transparent. Right (AU) puts a formant/chroma filter over the partials — vocal. Centre = neutral, 4 % deadzone.
- ENS

Ensemble / Drift — The partials follow the material a little more slowly and odd are detuned against even — an organic chorus that comes out of the sound itself.

5 SwarmPot 4

Granular texture — the cloudy, moving half of the blend.

- SIZ

Size — Grain length roughly 8-180 ms. Short = busy, long = smeared pad.
- SPR

Spread — Stereo width of the grains.
- SCN

Scan — 0 = freeze at one point in the buffer. Turn up = scrub through the take.
- DIR

Direction — Fwd · Rev · Rnd — playback direction per grain (Rnd = coin flip at spawn). The scan scrub runs independently.
- ATM

Atmosphere — Left Blur (BL): softer grain edges, an edgeless ambient cloud. Right Radiation (RD): lo-fi with tape saturation and warble.

With very dense clouds the module throttles the grain count by itself and shows L! in the header. Running grains finish their envelope — it never clicks, the cloud just thins out briefly.

6 ReverbPot 8

Space around the cloud — never around the clean input.

- MIX

Mix — How much reverb on the cloud.
- DEC

Decay — Reverb tail length.
- DMP

Damping — Higher = darker, softer tail.

6 · Reverb (continued)

- CHR

Character — Left Chorus (CH): slow modulation on the tail, wide and shimmering. Right Friction (FR): saturation inside the reverb itself, a dense overdrive wall at high values. Deliberately exclusive — one knob, two directions. Centre = plain.

7 ResonatorPot 7

Pitched ring on the Swarm side — eight modes over the granular cloud.

- MIX

Mix — How much resonant ring vs. dry Swarm.
- DEC

Decay — Ring time 0.08-8 s, exponential — short pluck to sustained hum. What used to be full scale now sits at the centre of the travel; the upper third is where it really sings.
- DMP

Damping — The wood-to-metal axis. At 0 all eight modes ring equally long — glassy, bell-like. Turned up, the upper modes die first, which reads as a struck wooden body.
- SPR

Spread — Fans the modes across the stereo field; the root stays centred, upper modes open outward. At 0 the bank is mono.
- PIT

Pitch — Root of the bank, ±1 octave around C2 ≈ 65.4 Hz.
- QNT

Quantized — OFF = odd harmonic series. ON = 8 scale degrees from Settings (SCL major/minor/pentatonic, INT equal or just).

Six entries — the segmented row scrolls.

8 Pan DriftPot 6

Trails wander left/right on their own.

- PHS

Phase — 0 % = all Trails drift in sync. 100 % = maximally offset, each on its own rhythm.
- AMP

Amplitude — How far L ↔ R. Zero = centred.
- VEL

Velocity — Speed and direction, bipolar up to ~2 Hz. Centre = still.

Moves the cloud only, never the clean input.

9 CrossfadePot 9

A loudness focus travels through the Trails.

- AMP

Amplitude — 0 % = all Trails equally loud. Higher = stronger spotlight on one Trail; the others are only attenuated and go quiet only near 100 %.
- VEL

Velocity — Travel speed; the side of centre sets the direction, centre = frozen focus (4 % deadzone).

Runs only across the Trails active per Count. Solo overrides the wave; Lock protects only against being overwritten, not against the wave. With Amplitude > 0, thin focus ticks travel beside T# on the dashboard.

10 Filter Pot 10

Tone shaping on a chosen stage of the cloud — never on the clean input.

- CUT
- Cutoff — Brightness of the filtered stage, 80 Hz to ~16 kHz.
- RES
- Resonance — Peak at the cutoff — whistle, bloom.
- FBK
- Feedback — Drive and growl into the filter.
- DST
- Destination — Where the filter sits: Off · Inp (both engines together, before the reverb) · Sp (Spectra only) · Sw (Swarm only) · Rv (on the reverb). Boot = Off.

TRAIL ENCODERS T1-T5

- **Turn** — Loudness of that Trail. Boot and reset default from Settings LVL (5 % steps, boot 50 %).
- **Short on Home** — Lock (L) — the Trail is neither overwritten nor does its hold time run out.
- **Short in a menu** — Back to the Trail overview. It does not lock — press short again on Home for that.
- **Long** — Solo (S), from any screen; ignores the Crossfade wave.

MULTI — BLOCK 11

- D/W
- Dry / Wet — Dry = clean input, Wet = the full cloud, cleanly blended in between. Boot ~55 %. Wet covers everything after capture: Spectra/Swarm with Resonator, filter inserts, reverb return, Pan Drift and Crossfade.
- M1 M2
- Macro 1 / 2 — Reserved — not wired yet.
- TU
- Time Unit — Reserved — seconds vs. clock, shared by Buffer and Hold.
- SET
- Settings — Opens the submenu on the right.

SETTINGS (SET)

- CPU
- CPU meter — Show or hide the load meter in the header.
- RAM
- RAM meter — Extra memory figure next to it. Default off.

- SCL
- Scale — 0 major · 1 minor · 2 pentatonic — for Resonator QNT.
- INT
- Intonation — Equal temperament ↔ just, with QNT on.
- REC
- Rec bar — Life-Bar style while recording: PLR (default) Perseid embers L→R, PRS the same embers centre→out, CTR solid centre→out. The Perseid styles soft-gate ~200 ms out before Fade In takes over.
- LVL
- Trail lvl — Default Trail level 0-100 % in 5 % steps (boot 50 %). Applies on boot, after delete-all and when edited — unlocked Trails only.
- #T
- Trail cnt — Default active Trail count 1-5 (boot 3). Also sets live CNT; the pot can still override it for the session without rewriting the default.

Planned but not yet operable: Instant Playback, auto-mod/normalling (Off · Age · Pitch · Both), audio routing stereo ↔ sidechain and FX→Input.

HOME DASHBOARD

- One row per active Trail: T# · level % · L (lock) · S (solo) · Life-Bar. The L/S columns stay reserved even when empty so the Life-Bar never jumps.
- Life-Bar phases: recording in the REC style from Settings, Fade In fills solid from the left, Hold shows the full bar with an inverted countdown or INF, Fade Out empties from the left, Empty is just an outline.
- On the left the input VU with the threshold marker — deliberately conservative, so room noise doesn't make the meter look hot.

PLAY IDEAS

Combinations spanning several blocks — all based on features that already exist.

- **Slowly ride Blend SP → SW** — Morph from harmonic-additive to grainy-granular over minutes — carries long ambient transitions without a hard cut.
- **Imprint on the highlight moment** — Press short exactly when it sounds good: freezes the active Trails while new takes layer on top.

- **Pitch Both up, PSP/PSW opposed** — One engine up, the other down — with PB widening both to ±2 octaves. Blending then smears the sound across octaves instead of switching cleanly.
- **Step through Filter Destination, cutoff fixed** — Leave cutoff and resonance alone, only switch DST (Inp → Sp → Sw → Rv) — the same filter curve feels completely different on each stage.
- **Threshold high + Overwrite OFF** — Only the loudest moments get captured at all, and then they stay — deliberately collects highlights instead of constantly overwriting.
- **Scan = 0 (freeze) + step through Lock/Solo** — Swarm freezes at one point in the buffer; step through the five Trails via Solo — browse frozen texture snapshots one at a time.
- **Resonator: decay upper third, damping 0, QNT on** — All eight modes ring equally long and sit on the scale from Settings — glassy bells over the granular cloud. Turning up Spread fans them into the stereo field.
- **Ride Reverb Character between Chorus and Friction** — Turn from lush shimmer on the left to a saturated wall on the right while a Trail decays — the reverb itself becomes the expressive tool.
- **High Pan Drift Phase + slow Crossfade** — Trails wander independently in space while a loudness focus slowly travels along — a sound picture that never repeats exactly.
- **Settings #T to 1-2, Hold to INF** — Deliberately few voices that stay put: every new take is a decision. Good for placing layers cleanly instead of letting them happen by accident.

P.S. Every Perseids-Rec pixel on the OLED — and each little flight path — was calculated by hand. ;)